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Investigations of harmonic generations in aperiodic optical superlattices

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We envisage the enhanced effect of harmonic generations from aperiodic optical superlattices (AOSs) achieved by inverting poled ferroelectric domains in samples. The search of optimal AOSs structures belongs to solving a difficult inverse source problem in nonlinear optics. We employ the simulated annealing method to successfully deal with this problem. We present the analyses of the design idea in the real-space representation and carry out several model designs. The constructed AOSs implement multiple wavelength second-harmonic generation and the coupled third-harmonic generation with an identical effective nonlinear coefficient at the preassigned wavelengths. The simulations show that the harmonic generations in the constructed AOSs can approach the prescribed goal better than those with the Fibonacci optical superlattice. The effective nonlinear coefficients versus the optical wave propagating distance from the impinging surface of incident light in the sample exhibit monotonically increasing behavior. This clearly infers that the contribution from each and every block to the optical parametric processes is in the constructive interference state. We also investigate the influence of the random fluctuation of the thickness of blocks on the performance of the constructed AOSs in detail for simulating practical fabrications and testing the performance stability of the sample.
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I. INTRODUCTION

With advances in the electric poling technique, it is possible to build various optical superlattices based on ferroelectric crystals such as $LiTaO_3$, $LiNbO_3$, $KTiOPO_4$, and so on, with periodically or quasiperiodically domain-inverted structures.[1–6] This development substantially extends materials available in the nonlinear optical area. In traditional nonlinear optics, in order to obtain high conversion efficiency in nonlinear optical processes, it usually requires the so-called phase-matching (PM) condition.[7,8] Normally, the PM may be achieved in nonlinear optical crystals with birefringence. This important PM condition gives rise to a large restriction to the choice of natural birefringence materials to serve in nonlinear optics. This PM method can only be suitable to some of the nonlinear processes in uniaxial or biaxial crystals. Another scheme, the so-called quasi-PM (QPM),[9,10] was proposed for a long time. This method substantially extends the class of materials available to various parametric interaction processes, including nonbirefringent crystals.[11–14]

QPM uses periodic modulation of the nonlinear property of a crystal to compensate for the mismatch between the wave vectors of the interacting light beams.[9,10] This allows utilization of the large component of the nonlinear susceptibility tensor, which is normally inaccessible with birefringent materials, and the implementation of noncritical QPM at any wavelength within a transparency range of crystals.

Unlike the periodic optical superlattice (POS), a one-dimensional quasi-POS (QPOS), for example, a Fibonacci optical superlattice (FOS)[1–3] with an infinite number of blocks provides a series of reciprocal vectors governed by two integers and the reciprocal vectors can be adjusted by changing structural parameters of the primitive block of the FOS. Therefore, some coupled optical parametric processes may be easily realized with high conversion efficiency in this kind of QPOSs. Some researchers have been devoted to theoretical and experimental studies on this subject for a long time.[1–3,11,12]

One question is raised whether this FOS is the best one for nonlinear optical processes. As is well known, a one-dimensional aperiodic optical superlattice (AOS) provides many more spatial Fourier spectra of structure than those of the FOS. Thus, it should be expected that the AOS may become another favorable candidate for optical parametric devices. The key question is how to realize the optimization design of the AOS for matching the specified nonlinear optical process with efficient conversion. This belongs to a relatively difficult inverse source problem in nonlinear optics. One believes that the extension of material class in the

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nonlinear optics area may follow a trace from homogeneous crystals, periodical structures (superlattices), and quasiperiodic structures to aperiodic structures.

The use of aperiodic QPM gratings for increasing the wavelength acceptance bandwidth in cw second harmonic generation (SHG) has been studied.[15–17] More recently, Arbore *et al.* proposed using chirped-period-poled lithium niobate gratings for QPM SHG to achieve pulse compression theoretically[18] and later they carried out the relevant experiment.[19]

Motivated by these works and questions, we have studied primarily the optimal design of the AOS and its performance.[20] In this paper we present extensive investigations of the optimization design method for constructing AOS with use of the simulated annealing (SA) method and the performance of the constructed AOS. We first give a description of the optimal design problem of the AOS in a real-space representation. We then carry out several model designs to demonstrate the effectiveness of the present design method. We also envisage the influence of random fluctuation of thickness of blocks on the performance of the constructed AOS in detail for simulating practical fabrications and testing the performance stability of samples.

The outline of this paper is as follows: Sec. II describes the necessary formulas used in calculations and the design method. The calculation results are presented in Sec. III with analyses. Finally, Sec. IV concludes with a brief summary.

II. THEORETICAL FORMULAS AND DESIGN METHOD

We consider an AOS based on $LiTaO_3$ (LT) crystals with laminar ferroelectric–domain structure. In such a material, the directions of polarization vectors in successive domains are opposite, as are the signs of nonlinear optical coefficients. However, the width of each individual domain may be different and it is determined by the specified optical parametric processes. This structure forms a one-dimensional modulation of nonlinear coefficient along the normal direction of domain layers in superlattices.

We first discuss the SHG process for the single wavelength case. Assume that a laser beam with $\omega_1=\omega$ is perpendicularly incident from the left onto the surface of an AOS, and through the nonlinear optical process, the SHG is generated in this AOS. In order to use the largest nonlinear coefficient d_{33}, let the interfaces of each domain be parallel to the yz plane, and the propagation and the polarization directions of incident light are along the x and z axes, respectively. In the SHG, two optical fields are involved: one for fundamental wave $\omega_1=\omega$, and other for the second-harmonic wave $\omega_2=2\omega$. The electric field of the fundamental wave with ω results in nonlinear polarization at frequency 2ω, which leads to the SHG with double fundamental frequency. In the assumption of slowing-wave variation of field amplitudes and the so-called small-signal approximation in which the depletion loss of fundamental wave power can be negligible, the conversion efficiency η_{SHG} from fundamental wave to second-harmonic wave reads

$$\eta_{\mathrm{SHG}}=\frac{I_{2\omega}}{I_\omega}=\frac{8\pi^2|d_{33}|^2 I_\omega L^2}{c\epsilon_0\lambda^2 n_{2\omega}n_\omega^2}\left|\frac{1}{L}\int_0^L dx\, e^{i(k_{2\omega}-2k_\omega)x}\tilde{d}(x)\right|^2, \tag{1}$$

where $k_\omega=n_\omega\omega/c$ ($k_{2\omega}=2n_{2\omega}\omega/c$) is the wave number of the fundamental (second harmonic) wave, c is the speed of light in vacuum, n_ω ($n_{2\omega}$) represents the refractive index of material at fundamental (second harmonic) wavelength, λ is the fundamental wavelength, ϵ_0 is the permittivity of vacuum, and L is the total length of the sample. $I_\omega(I_{2\omega})$ is the intensity of the fundamental (second harmonic) wave beam, and $\tilde{d}(x)$ only takes binary values of 1 or -1. The coherence length $l_c^{(s)}(\lambda)$ for the SHG is defined by[8]

$$l_c^{(s)}(\lambda)=\frac{2\pi}{\Delta k}=\frac{2\pi}{k_{2\omega}-2k_\omega}=\frac{\lambda}{2(n_{2\omega}-n_\omega)}. \tag{2}$$

By introducing this quantity, Eq. (1) can be rewritten as

$$\eta_{\mathrm{SHG}}=C_s C^2(\lambda)\,\xi_{\mathrm{eff}}^{(s)2}(\lambda), \tag{3}$$

where

$$C_s=\frac{8\pi^2|d_{33}|^2 I_\omega L^2}{c\epsilon_0}, \tag{4}$$

$$C(\lambda)=\frac{1}{\lambda\sqrt{n_{2\omega}n_\omega}}, \tag{5}$$

$$\xi_{\mathrm{eff}}^{(s)}(\lambda)=\left|\frac{1}{L}\int_0^L dx\, e^{i(2\pi x/l_c^{(s)}(\lambda))}\tilde{d}(x)\right|, \tag{6}$$

where $\xi_{\mathrm{eff}}^{(s)}(\lambda)$ stands for the reduced effective nonlinear coefficient of the SHG.

We now discuss the inverse source problem in constructing the optical AOS. Assume that the thickness of the unit block is Δx, thus the number of blocks in the sample is $N=L/\Delta x$. The position of each block is given by $x_q=q\Delta x$, for $q=0,1,2,3\ldots(N-1)$. We evaluate the integral Eq. (6) as

$$\begin{aligned}\xi_{\mathrm{eff}}^{(s)}(\lambda)&=\frac{1}{N\Delta x}\left|\sum_{q=0}^{N-1}\tilde{d}(x_q)\int_{x_q}^{x_q+\Delta x}d\zeta\, e^{i(2\pi\zeta/l_c^{(s)}(\lambda))}\right|\\&=\left|\mathrm{sinc}\left[\frac{\Delta x}{l_c^{(s)}(\lambda)}\right]\right|\\&\quad\times\left|\left\{\frac{1}{N}\sum_{q=0}^{N-1}\tilde{d}(x_q)e^{i[2\pi(q+0.5)\Delta x/l_c^{(s)}(\lambda)]}\right\}\right|,\end{aligned} \tag{7}$$

where $\mathrm{sinc}(x)=\sin(\pi x)/(\pi x)$. It is clearly seen that $\xi_{\mathrm{eff}}^{(s)}(\lambda)$ has contributions from two factors: one factor (sinc function) belongs to the unit block and strongly depends on Δx and the coherence length $l_c^{(s)}(\lambda)$; the other factor, which is included inside the curly braces, reflects the interference effect among the blocks and depends on the configuration of domains as well as the phase lagging from one block to the other.

It is evident that the optimization design of the AOS for the SHG can be ascribed as a search for the maximum of $\xi_{\mathrm{eff}}^{(s)}(\lambda)$ with respect to $\tilde{d}(q\Delta x)$. For the first factor in Eq. (7), this maximum value condition demands $\Delta x=[(2m+1)/2]l_c^{(s)}(\lambda)$ for $m=0,1,2,\ldots$, and the maximum values of this factor are $2/[\pi(2m+1)]$. The maximum value condition for the second factor corresponds to the appearance of the perfectly constructive interference. In the case of the periodic structure with a period of $a=2\Delta x$, $d(x_q)$ takes opposite sign between two consecutive blocks, i.e., $\tilde{d}(q\Delta x)=(-1)^q$, for $q=0,1,2\ldots(N-1)$, we obtain

$$\text{the second factor}=\frac{1}{N}\left|\sum_{q=0}^{N-1}(-1)^q e^{i\pi(2m+1)(q+0.5)}\right|=\frac{1}{N}\left|i(-1)^m\sum_{q=0}^{N-1}(-1)^{2q}\right|=1. \quad (8)$$

In this case, the phase lagging factor perfectly matches the reversal of the domain orientation between two adjacent blocks and the ideal constructive interference emerges. However, in the case of the aperiodic structure, the situation becomes much more complicated, and it belongs to solving an inverse source problem. Such an optimization nonlinear problem can be solved with the SA method[21,22] in which an objective function E should be minimized, and then the favorable arrangement of the domain orientations of the blocks in the sample can be determined fully.

To demonstrate the effectiveness and usefulness of the SA method for dealing with the inverse source problem concerned here, we consider a simple example of a single wavelength SHG to verify the SA method available. We choose the parameters: the wavelength of incident beam is $\lambda=1.064$ μm, the thickness of the unit block is $\Delta x=l_c^{(s)}/2=3.8713$ μm, and the number of blocks is 2142. Thus, the total length of sample is $L=8298.8$ μm. The refractive index of material at fundamental (second harmonic) wavelength is evaluated by the Sellmeier equation given by Ref. 4. The objective function in the SA is chosen as

$$E=|\xi^{(0)}-\xi_{\text{eff}}^{(s)}(\lambda)|, \quad (9)$$

where $\xi^{(0)}$ is a predesignated value in guiding the SA procedure. Finally, we obtain a perfect periodical structure with a pair of antiparallel domains for each unit block and with a period of $a=2\Delta x=l_c^{(s)}$, as expected. The reduced effective nonlinear coefficient $\xi_{\text{eff}}^{(s)}$ reaches its theoretical maximum value of $2/\pi=0.6366$. This means that the SA method is appropriate for dealing with the above-mentioned inverse source problem.

It is worth emphasizing that the above result has the scaling invariance for any sum-frequency process, including SHG. In the sum-frequency generation (SFG), the corresponding expression of the effective nonlinear coefficient is the same as before, except that the coherence length is replaced by the corresponding new one.

III. RESULTS AND ANALYSES

A. Multiple wavelength SHG

We now carry out a particular design of the AOS that implements multiple wavelength SHG with an identical effective nonlinear coefficients $\xi_{\text{eff}}^{(s)}(\lambda_\alpha)=\xi^{(0)}$, where $\xi^{(0)}$ is a constant. First, we have to carefully choose an appropriate thickness Δx of the unit block. Δx is mainly dominated by the first factor in Eq. (7). This factor possesses a strong dispersion due to the strong wavelength dependence of $l_c^{(s)}(\lambda_\alpha)$. As is well known, $\text{sinc}(\zeta)\sim 1$ when ζ is small, thus the dispersion can be negligible. Therefore, we choose $\Delta x=3$ μm for matching the state of the art of microfabrication in practice. The objective function in the SA method is selected as

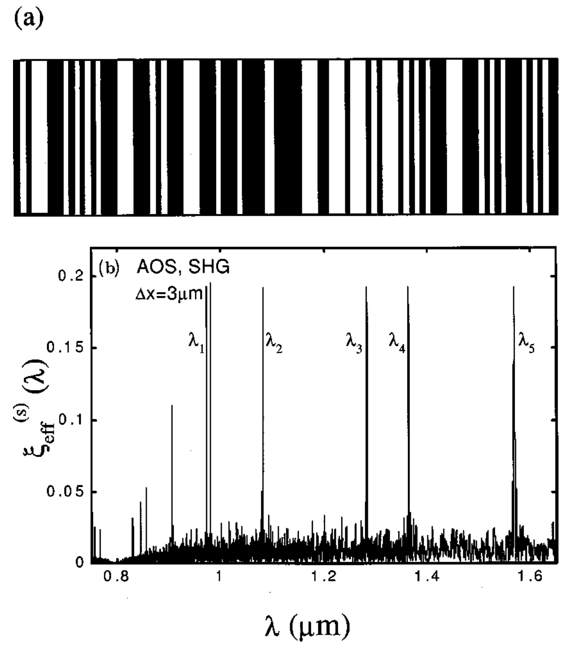

FIG. 1. Calculated results for the constructed AOS that implements multiple wavelength SHG with an identical nonlinear optical coefficient $\Delta x=3$ μm: (a) a gray-scale diagram of the constructed AOS in part; (b) dependence of $\xi_{\text{eff}}^{(s)}(\lambda)$ of the AOS on wavelength in a sampling wavelength interval of 0.5 Å.

$$E=\sum_\alpha [|\xi^{(0)}-\xi_{\text{eff}}^{(s)}(\lambda_\alpha)|]+\beta[\max\{\xi_{\text{eff}}^{(s)}(\lambda_\alpha)\}-\min\{\xi_{\text{eff}}^{(s)}(\lambda_\alpha)\}], \quad (10)$$

where the symbol $\max\{\cdots\}$ ($\min\{\cdots\}$) manifests to take their maximum (minimum) value among all the quantities including into $\{\cdots\}$. β is an adjustable parameter, taking a value of 0.3–3.0. λ_α takes the following values: 0.972 00, 1.082 00, 1.283 00, 1.364 00, and 1.568 70 (μm). The reason for choosing these wavelengths in the design is a convenience for a comparison of our result with that in Ref. 2, in which the sample is a FOS of LT. The dispersion relation of the refractive index of LT crystal at fundamental and second harmonic wavelengths is calculated as $T=25\,°C$ according to the Sellmeier formula given in Ref. 4.

We adopt the parameters in the calculations as: the total length of sample $L=8295$ μm, the number of blocks $N=2765$, and the number of sampling wavelengths is 800 within the range of [0.85, 1.65] μm, i.e., the sampling wavelength interval is 10 Å. The calculation results are displayed

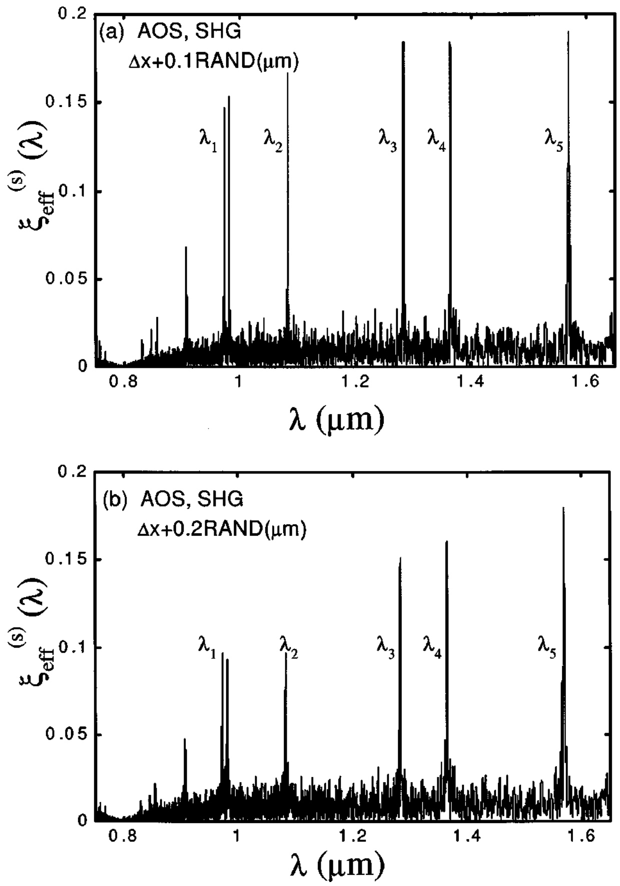

FIG. 2. Influence of random fluctuation of the thickness of blocks in the AOS sample given in Fig. 1 on $\xi_{\text{eff}}^{(s)}(\lambda)$. Random function with a uniform density distribution within the range of $[-1,1]$ is taken into account, referred as RAND: (a) for $\Delta x+0.1$RAND (μm) and (b) for $\Delta x+0.2$RAND (μm). $\Delta x=3$ μm.

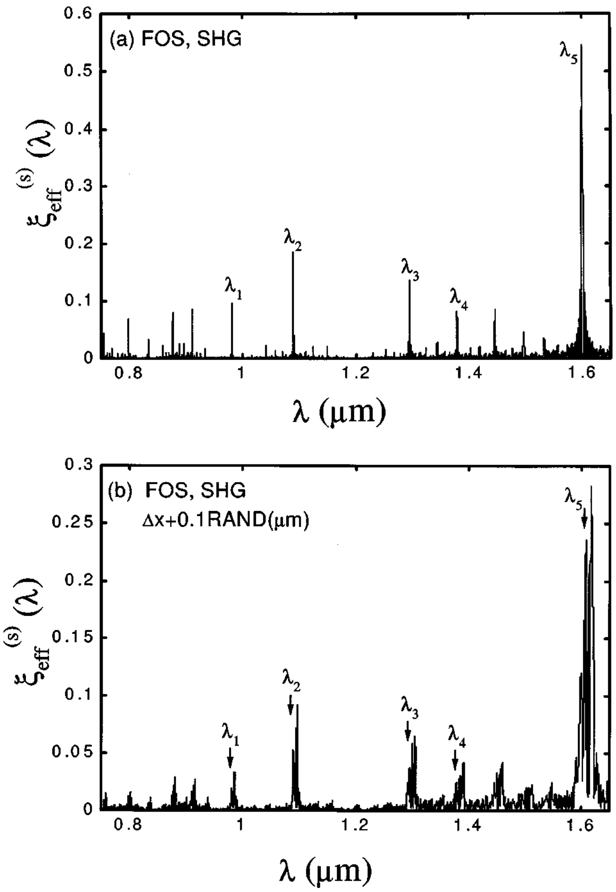

FIG. 3. (a) Dependence of $\xi_{\text{eff}}^{(s)}(\lambda)$ on λ for the Fibonacci optical superlattice (FOS) presented in Ref. 2. (b) Stability of the performance of this sample under the fluctuation of the thickness of blocks. The fluctuation strength is 0.1RAND (μm).

in Fig. 1. Figure 1(a) is a gray-scale diagram of the constructed AOS in part. The black (white) strip represents the positive (negative) domain. Figure 1(b) displays the calculated reduced effective nonlinear coefficient as a function of wavelength for the SHG after scanning a wide range of wavelengths and with a sampling wavelength interval of 0.5 Å, much smaller than that in the design procedure. There exist six strong peaks with almost an identical peak value. Five of them are located at the preset wavelengths. One strong peak with an unexpected wavelength $\lambda=0.980\,95$ μm appears very close to the excepted wavelength $\lambda=0.972\,00$ μm. There are a few stray peaks appearing in the lower wavelength regions and small dense oscillation structures as a background of the whole curve. One of the stray peaks with a medium strength is located at $\lambda=0.906\,75$ μm. The average value of $\langle\xi_{\text{eff}}^{(s)}(\lambda_\alpha\rangle$ for these five main peaks is 0.1927 and the nonuniformity defined by $\Delta\xi_{\text{eff}}^{(s)}=1/\alpha[\Sigma_\alpha|\xi_{\text{eff}}^{(s)}(\lambda_\alpha)-\langle\xi_{\text{eff}}^{(s)}\rangle|]/\langle\xi_{\text{eff}}^{(s)}\rangle$ equals 3.18×10^{-4}.

To match the practical fabrication procedure and test the stability of the performance of sample, it is interesting to study the influence of random fluctuation in the thickness of blocks on the SHG. The variation of the effective nonlinear optical coefficient for SHG versus wavelength for different fluctuations of the thickness of blocks in the same sample as Fig. 1 is depicted in Fig. 2. Using the random distribution function with a uniform density in the range of $[-1,\,1]$ simulates the thickness fluctuation of blocks, referring to it as RAND. The strengths of fluctuation in Fig. 2 are: (a) for $\Delta x+0.1$ RAND (μm) and (b) for $\Delta x+0.2$ RAND (μm). The width of the unit block is $\Delta x=3$ μm. From these plots with a sampling wavelength interval of 0.5 Å, one observes that the peak positions are shifted slightly toward the high wavelengths from the preassigned wavelengths. As fluctuation strength increases, the average value of $\xi_{\text{eff}}(\lambda_\alpha)$ is decreased and nonuniformity is increased.

It is interesting to present a comparison of our constructed AOS with the FOS presented in Ref. 2. Employing the parameters in Ref. 2, we form the modulation function of $d_{33}(x)$ and repeat the calculation of Eq. (7), the dependence of $\xi_{\text{eff}}^{(s)}(\lambda)$ on λ now is depicted in Fig. 3(a). The profile of $\xi_{\text{eff}}^{(s)}(\lambda)$ consists of a series of peaks and the number of unequal height main peaks is much lager than five. The largest peak with $\xi_{\text{eff}}^{(s)}=0.5458$ is located at $\lambda=1.599\,90$ μm. The positions of the main peaks in $\xi_{\text{eff}}^{(s)}(\lambda)$ curve are substantially deviated from the prescribed ones and they are located, respectively, at the wavelengths of $[0.980\,55,\ 1.089\,30,\ 1.294\,50,\ 1.378\,45,\ 1.599\,90]$ μm after the search with use of a sampling wavelength interval of 0.5 Å. The wavelength deviations are $[+85.5,\ +73.0,\ +115.0,\ +120.5,\ +312.0,\ ]$Å from the preset values. The average value $\langle\xi_{\text{eff}}^{(s)}\rangle$ is 0.2098,

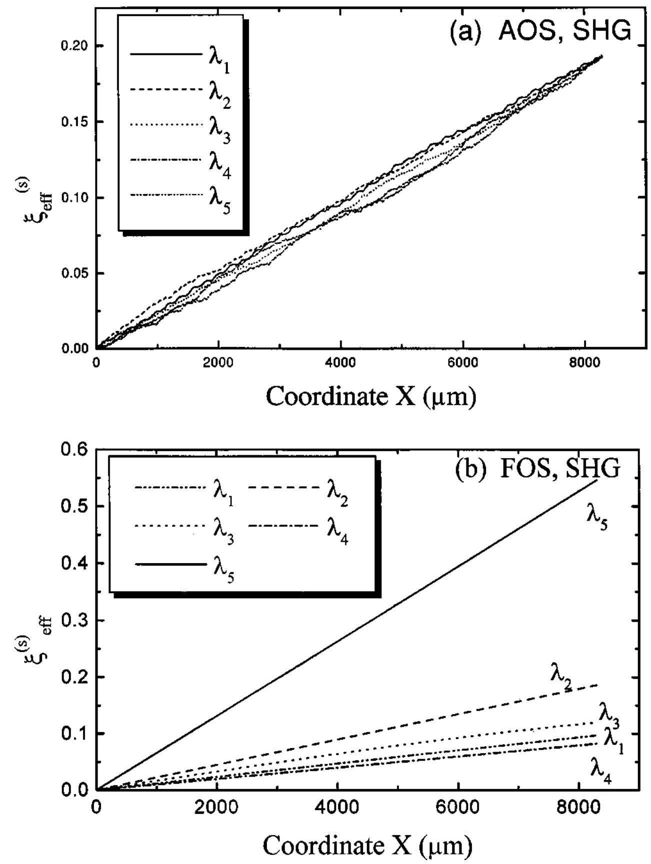

FIG. 4. (a) Variation of $\xi_{\rm eff}^{(s)}(\lambda)$ with the propagating distance of optical wave in the AOS sample given in Fig. 1 for five main strong peak wavelengths. (b) Variation of $\xi_{\rm eff}^{(s)}(\lambda)$ with the propagating distance of optical wave in the FOS sample given in Ref. 2 for five prescribed peak wavelengths.

quite close to our result of 0.1927. However, the behavior of $\xi_{\rm eff}^{(s)}(\lambda)$ exhibits significant nonuniformity for these five wanted peaks in Fig. 3(a). The nonuniformity reaches 64.44%, much larger than our result in the AOS above (3.18×10^{-2}%). On the other hand, a variety of unwanted stray peaks with the medium strengths take place in Fig. 3(a).

We also investigate the stability of the performance of this FOS to the fluctuation of thickness of poled blocks. The calculated result is shown in Fig. 3(b) for the case of $\Delta x + 0.1\text{RAND}$ (μm). It is evident that all five peaks [marked by arrows in the $\xi_{\rm eff}(\lambda_\alpha)$ plot] become much broader and contain spikes. In particular, the strongest one at $\lambda = 0.159\,990$ μm nearly develops into a miniband with spikes. Therefore, the pattern of $\xi_{\rm eff}^{(s)}(\lambda)$ is considerably distorted compared with the originally constructed FOS sample. It indicates that the performance stability of the FOS with respect to the thickness variance of blocks is much poorer than that of the AOS.

In order to further reveal the characteristic of the SHG in the constructed AOS, we present the plot of the variation of $\xi_{\rm eff}^{(s)}(\lambda_\alpha)$ with the optical wave propagating distance x from the impinging surface of incident light in Fig. 4(a), for five main peak wavelengths of Fig. 1. It is obviously seen that all curves exhibit nearly linearly increasing behavior with a nearly identical slope. It hints that the arrangement of domains is relatively favorable to the SHG process because the contribution from the successive blocks to $\xi_{\rm eff}(\lambda_\alpha)$ always takes positive one. The individual contribution is accumulated with each in the constructive interference state. This is the reason that the SHG in the AOS possesses large $\xi_{\rm eff}^{(s)}(\lambda)$ while having good uniformity at the desired wavelengths. For comparison, we also demonstrate the corresponding plot for the FOS presented in Ref. 2, as shown in Fig. 4(b). It is seen that all the curves also exhibit linearly increasing behavior, but with much different slopes. This is the reason that $\xi_{\rm eff}^{(s)}(\lambda)$ exhibits a strong dispersion behavior and leads to significant wavelength dependence of conversion efficiency of the SHG in the FOS sample.

B. The coupled THG for multiple wavelengths

Third harmonic generation (THG) has a wide application as a mean to extend coherent light sources to the short wavelengths. THG is created in several different ways. For instance, first, it is directly produced from a third-order nonlinear process which is of little practical importance because of intrinsic weak third-order optical nonlinearity; second, an efficient THG was achieved by a two-step process separately. Two nonlinear optical crystals are involved: the first one is for SHG and the other one is for SFG;[8] third, it may be generated from the coupled parametric processes with high efficiency.

The coupled THG (CTHG) is raised from the coupling effect of two nonlinear optical processes : one is the SHG and the other is a sum-frequency process. Hence the two processes no longer proceed alone but couple with each other. This coupling leads to a continuous energy transfer from the fundamental to the second, to the third-harmonic fields; thus, a direct third-harmonic wave was generated from the AOS with high efficiency.

The THG process is analyzed by solving the coupled nonlinear equations that describe interaction of these three fields: E_ω, $E_{2\omega}$, and $E_{3\omega}$ in the AOS. Under the small signal approximation, the third-harmonic-wave conversion efficiency $\eta_{\rm THG}^{(t)}$ is expressed by

$$\eta_{\rm THG} = \frac{I_{3\omega}}{I_\omega} = \frac{144\pi^4 |d_{33}|^4 I_\omega^2 L^4}{c^2 \epsilon_0^2 \lambda^4 n_\omega^3 n_{2\omega}^2 n_{3\omega}} \times \left| \frac{2}{L^2} \int_0^L dx\, e^{i(k_{3\omega}-k_{2\omega}-k_\omega)x} \tilde{d}(x) \times \int_0^x d\zeta\, e^{i(k_{2\omega}-2k_\omega)\zeta} \tilde{d}(\zeta) \right|^2 . \tag{11}$$

The coherence length $l_c^{(t)}(\lambda)$ for the CTHG is defined as

$$l_c^{(t)}(\lambda) = \frac{2\pi}{k_{3\omega}-k_{2\omega}-k_\omega} = \frac{\lambda}{(3n_{3\omega}-2n_{2\omega}-n_\omega)}. \tag{12}$$

By using this quantity and the coherence length of the SHG,

$l_c^{(s)}(\lambda)$ [given in Eq. (2)], Eq. (11) can then be rewritten as

$$\eta_{\mathrm{THG}}=C_t C'^2(\lambda)(\xi_{\mathrm{eff}}^{(ct)}(\lambda))^2, \tag{13}$$

where

$$C_t=\frac{144\pi^4|d_{33}|^4 I_\omega^2 L^4}{c^2\epsilon_0^2}, \tag{14}$$

$$C'(\lambda)=\frac{1}{\lambda^2 n_\omega^{3/2} n_{2\omega}\sqrt{n_{3\omega}}}, \tag{15}$$

$$\xi_{\mathrm{eff}}^{(ct)}(\lambda)=\left|\frac{2}{L^2}\int_0^L dx\, e^{i[2\pi x/l_c^{(t)}(\lambda)]}\tilde{d}(x)\times\int_0^x d\zeta\, e^{i[2\pi\zeta/l_c^{(s)}(\lambda)]}\tilde{d}(\zeta)\right|, \tag{16}$$

where $\xi_{\mathrm{eff}}^{(ct)}(\lambda)$ stands for the reduced coupled effective nonlinear coefficient for the CTHG.

We now discuss the optimization design problem of the AOS structure. Assume that the thickness of the unit block is Δx, thus the number of the blocks in the sample is $N=L/\Delta x$. The position of the blocks is coordinated with $x_q=q\Delta x$, for $q=0,1,2,3\ldots(N-1)$. Evaluating the integral, Eq. (16), we obtain

$$\begin{aligned}\xi_{\mathrm{eff}}^{(ct)}(\lambda)&=\left|\frac{2}{(N\Delta x)^2}\sum_{q=0}^{N-1}\tilde{d}(x_q)\int_{x_q}^{x_q+\Delta x}dx\, e^{i[2\pi x/l_c^{(t)}(\lambda)]}\sum_{p=0}^{q}\tilde{d}(x_p)\int_{x_p}^{x_p+\Delta x}d\zeta\, e^{i[2\pi\zeta/l_c^{(s)}(\lambda)]}\right|\\
&=2\left|\left[\frac{1}{\Delta x}\int_0^{\Delta x}dx e^{i[2\pi x/l_c^{(t)}(\lambda)]}\right]\left[\frac{1}{\Delta x}\int_0^{\Delta x}d\zeta e^{i[2\pi\zeta/l_c^{(s)}(\lambda)]}\right]\right.\\
&\quad\left.\times\left\{\frac{1}{N}\sum_{q=0}^{N-1}\tilde{d}(x_q)e^{i[2\pi x_q/l_c^{(t)}(\lambda)]}\left[\frac{1}{N}\sum_{p=0}^{q-1}\tilde{d}(x_p)e^{i[2\pi x_p/l_c^{(s)}(\lambda)]}\right]\right\}+\Delta U\right|\\
&=2\left|\left\{\mathrm{sinc}\left[\frac{\pi\Delta x}{l_c^{(t)}(\lambda)}\right]\mathrm{sinc}\left[\frac{\pi\Delta x}{l_c^{(s)}(\lambda)}\right]\right\}\left\{\frac{1}{N^2}\sum_{q=0}^{N-1}\tilde{d}(x_q)e^{i[2\pi(q+0.5)\Delta x/l_c^{(t)}(\lambda)]}\left[\sum_{p=0}^{q-1}\tilde{d}(x_p)e^{i[2\pi(p+0.5)\Delta x/l_c^{(s)}(\lambda)]}\right]\right\}+\Delta U\right|,\end{aligned} \tag{17}$$

where

$$\begin{aligned}\Delta U&=\frac{2}{L^2}\sum_{q=0}^{N-1}\tilde{d}(x_q)\int_{x_q}^{x_{q+1}}dx\, e^{i[2\pi x_q/l_c^{(t)}(\lambda)]}\\
&\quad\times\int_{x_q}^{x}d\zeta\, e^{i[2\pi\zeta/l_c^{(s)}(\lambda)]}\\
&=\frac{1}{N^2}\left(\frac{l_c^{(s)}(\lambda)}{i\pi\Delta x}\right)\{e^{i[\pi\Delta x(1/l_c^{(s)}(\lambda)+1/l_c^{(t)}(\lambda))]}\\
&\quad\times\mathrm{sinc}[\Delta x(1/l_c^{(s)}(\lambda)+1/l_c^{(t)}(\lambda)]\\
&\quad-e^{i[\pi\Delta x/l_c^{(t)}(\lambda)]}\mathrm{sinc}[\Delta x/l_c^{(t)}(\lambda)]\}\\
&\quad\times\sum_{q=0}^{N-1}e^{i[2\pi x_q(1/l_c^{(s)}(\lambda)+1/l_c^{(t)}(\lambda))]}.\end{aligned} \tag{18}$$

Thus, $\xi_{\mathrm{eff}}^{(ct)}(\lambda)$ in Eq. (17) has the contributions from two factors: one factor (double sinc functions) belongs to the unit block and strongly depends on Δx and the coherence length $l_c^{(t,s)}(\lambda)$; the other factor, which is included inside the second set of curly braces, reflects the interference effect among the blocks in the sample, depending on the arrangement of domains and the phase lagging from one block to the other. ΔU contributes to a small correction.

We now introduce the following three effective nonlinear coefficients for the convenience of discussions: The effective nonlinear coefficient $\xi_{\mathrm{eff}}^{(s)}(\lambda)$ with $l_c^{(s)}(\lambda)$ for SHG, the effective SFG coefficient $\xi_{\mathrm{eff}}^{(t)}(\lambda)$ with $l_c^{(t)}(\lambda)$ for THG, and the coupled effective nonlinear coefficient $\xi_{\mathrm{eff}}^{(ct)}(\lambda)$ for CTHG. Their detailed expressions are as follows: The effective nonlinear coefficient for SHG is given by Eq. (7); the effective sum-frequency generation coefficient for THG is given by a similar form in Eq. (7) except for replacing all $l_c^{(s)}(\lambda)$ by $l_c^{(t)}(\lambda)$. The coupled effective nonlinear coefficient for THG is given in Eqs. (17) and (18).

Compared with the two-step processes, the conversion efficiency in an AOS is increased by a factor of

$$\frac{(\xi_{\mathrm{eff}}^{(ct)})^2 L^4}{(\xi_{\mathrm{eff}}^{(s)}\xi_{\mathrm{eff}}^{(t)})^2(0.5L)^2(0.5L)^2}. \tag{19}$$

The reason for this is that in the traditional two separate steps scheme, the second-harmonic and sum-frequency processes are not coupled with each other. In order to present a fair comparison, the length of each sample in the two-step scheme should only take one half of the total length of the AOS sample in which two processes are coupled.

We now carry out the model design of the AOS that achieves the CTHG. The parameters in the design are: $\Delta x=3$ μm, $L=8067$ μm, $N=2689$, and $\lambda=1.570$ μm. Figure

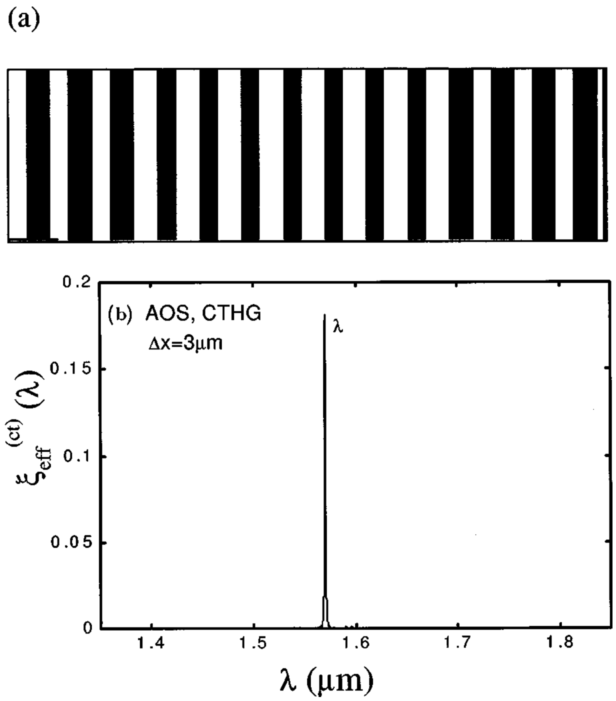

FIG. 5. Calculated results for the constructed AOS that implements the coupled THG with an identical nonlinear coefficient for a single wavelength $\Delta x=3$ μm: (a) a gray-scale diagram of the constructed AOS in part; (b) $\xi_{\rm eff}^{(ct)}(\lambda)$ vs λ with the use of a sampling wavelength interval of 1.0 Å.

5 displays the calculated results. Figure 5(a) is a gray-scale diagram of the constructed AOS in part. It is clearly observable that the arrangement of poled domains in the sample is no longer a periodic structure, completely different from the case of the SHG with a single wavelength. Figure 5(b) displays the reduced effective nonlinear coefficient $\xi_{\rm eff}^{(ct)}(\lambda)$ of the constructed AOS after scanning a wide range of wavelengths and with a sampling wavelength interval of 1.0 Å. There exists only one strong sharp peak at the predesignated wavelength of $\lambda=1.570$ μm, $\xi_{\rm eff}^{(ct)}=0.1811$. For the SHG alone and the SFG alone, the ideal maximum value of $\xi_{\rm eff}^{(s,t)}$ should be 0.6366 Thus, compared with the two-step process, the conversion efficiency of the CTHG in an AOS is increased by a factor of 3.1915, evaluated by Eq. (19).

We now show the characteristics of the constructed AOS that implements multiple wavelength coupled THG with an identical effective nonlinear coefficient. The relevant parameters are: $\lambda_\alpha=[1.40, 1.60, 1.80]$ μm, $\Delta x=3\,\mu$m, $L=8067$ μm, and $N=2689$. The dependences of $\xi_{\rm eff}^{(s,t,ct)}(\lambda)$ on the wavelength are depicted in Figs. 6(a)–6(c), respectively. There exist three peaks located at the prescribed wavelengths. However, in Fig. 6(b), there is one extra unexpected peak appearing at $\lambda=1.4435$ μm. In Figs. 6(a) and 6(c), one can observe noisy background, but the noise in Fig. 6(c) is strongly suppressed to a negligible small level, and the extra peak becomes indistinguishable. $\xi_{\rm eff}^{(ct)}(\lambda)$ exhibits a fairly good uniformity with an average value of 0.047 92.

In order to further show the characteristic of the CTHG in the constructed AOS, we plot the variation of $\xi_{\rm eff}^{(s,t,ct)}(\lambda_\alpha)$ with the optical wave propagating distance x from the impinging surface of the incident light in Fig. 7, for three main

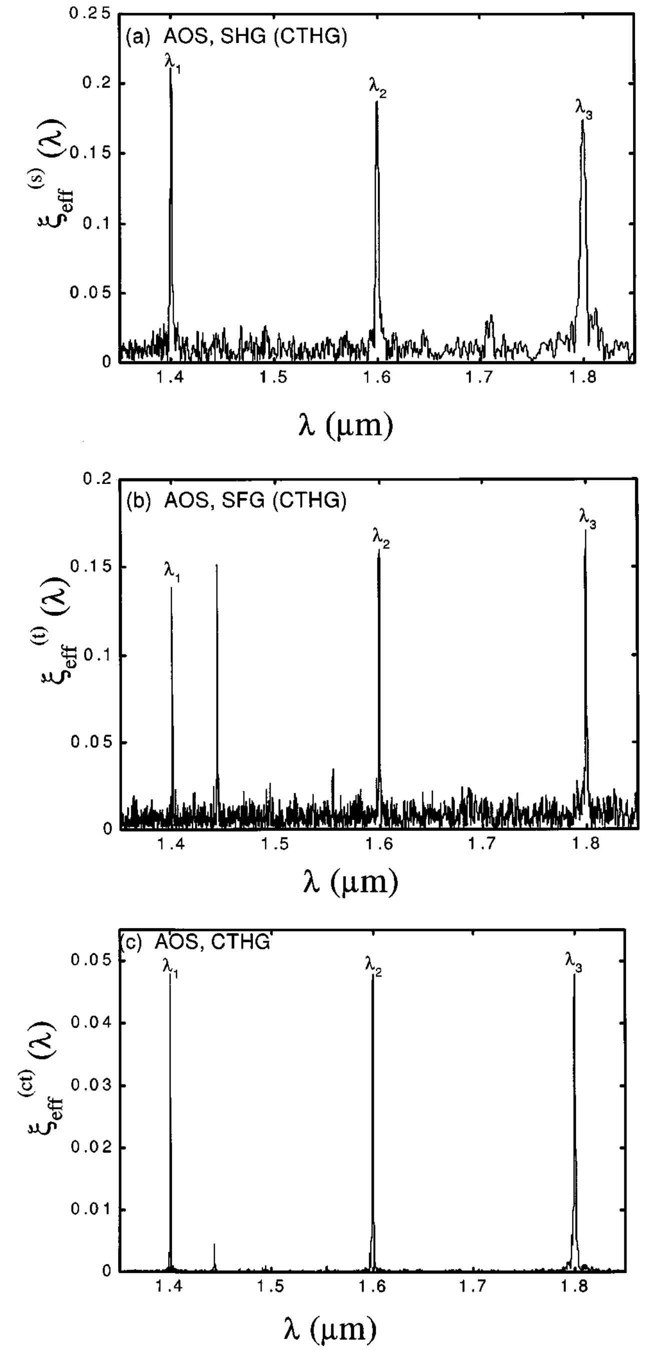

FIG. 6. Calculated results for the constructed AOS that implements the CTHG with an identical nonlinear coefficient for three wavelengths $\Delta x=3$ μm: (a) $\xi_{\rm eff}^{(s)}(\lambda)$ vs λ; (b) $\xi_{\rm eff}^{(t)}(\lambda)$ vs λ; and (c) $\xi_{\rm eff}^{(ct)}(\lambda)$ vs λ with use of a sampling wavelength interval of 1.0 Å.

peak wavelengths of Fig. 6. It is obviously seen that they exhibit monotonically increasing grouped curves with different rising slopes. It hints that the arrangement of the domains is relatively favorable to the SHG process because the contribution from every block to $\xi_{\rm eff}^{(ct)}(\lambda)$ always takes a positive one.

It is interesting to indicate the difference of the performances between the AOS and the FOS for the CTHG process. Recently Zhu *et al.* have reported the experimental and theoretical results of the CTHG of the FOSs in detail.[1] They designed a FOS made from a single crystal with a quasiperi-

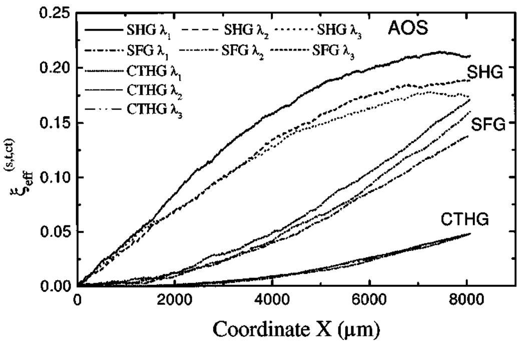

FIG. 7. Variation of $\xi_{\rm eff}^{(s,t,ct)}(\lambda)$ with the propagating distance of optical wave in the AOS sample given in Fig. 6 for three main strong peak wavelengths.

odic laminar ferroelectric domain structure to realize a CTHG process.[1] It is expected that a FOS can provide more plentiful reciprocal-lattice vectors to compensate the mismatch phase in optical parametric process, which causes most of third-harmonic peaks labeled by the corresponding reciprocal-lattice vector indices. However, numerical calculations show that some CTHG peaks with rather strong intensity cannot be labeled with the relevant reciprocal-lattice vector indices of the FOS. Even for the THG peaks labeled only by THG indices, most of the THG peaks still occur without the corresponding SHG peak. This result can be understood well because the situations of the THG QPM can be usually met, whereas a simultaneous appearance of QPM in both SHG and THG rarely occurs by changing the geometric structural parameter of the FOS. It gives rise to the reduction of the conversion efficiency of the CTHG because CTHG is associated with two parametric processes: the SHG and the SFG processes. The strong CTHG peaks demand the satisfaction of the QPM in both SHG and SFG processes, leading to a THG growth as L^4, where L is the length of the FOS sample. They also demonstrated both plots of the dependence of the CTHG and SHG intensities on block numbers within the FOS. They found that the labeled or unlabeled third-harmonic intensity is usually grown with a block number and behaves as an increasing curve in a stepwise fashion, whereas the involving mediated second-harmonic intensity oscillates periodically. In this case, the SHG process is severely quasiphase-mismatched, but the THG process may be either quasiphase-matched or quasiphase-mismatched. Each plateau in the third-harmonic intensity curve is just located at the vicinity of a valley of the related SHG intensity oscillatory curve. It concludes that the constructed FOS is far away from the preassigned goal.

It is worth pointing out that our constructed AOS always made all peaks in $\xi_{\rm eff}^{(s,t,ct)}(\lambda)$ spectra appearing at the prescribed wavelengths without stray peaks, except for one extra peak in $\xi_{\rm eff}^{(t)}(\lambda)$. However, this extra peak does not have any active effect on $\xi_{\rm eff}^{(ct)}(\lambda)$, in which only three strong peaks take place at the predesignated wavelengths, as seen in Fig. 6(c). It also infers that the QPM condition is well met simultaneously for both the SHG and SFG processes in the CTHG of the AOS sample.

IV. SUMMARY

We have presented an extensive investigation of the characteristics of harmonic generations from the AOS. The optimal design of the AOS for the harmonic generation belongs to a difficult inverse source problem in nonlinear optics. We propose to use the SA method to deal with this problem. We describe the design idea and carry out several model designs. For instance, the constructed AOS performs multiple SHG or the CTHG with an identical effective nonlinear coefficient at the prescribed wavelengths. All the simulation results demonstrate that the constructed AOS possesses the performance better than that of the FOS presented in Ref. 2 with the respects of matching accurately the preset wavelengths of the incident light and the uniformity of the peak value of $\xi_{\rm eff}^{(s,ct)}(\lambda)$. We show the variations of nonlinear optical coefficients with the optical wave propagating distance from the impinging surface of the incident light beam in the AOS sample and find that they demonstrate a monotonically increasing behavior with a positive rising slope. This clearly manifests that the contribution from individual block to optical parametric processes is in the constructive interference state. We also survey the performance stability of the AOS for the harmonic generations by examining the variation of harmonic generation efficiency with respect to the fluctuation of the thickness of blocks. We find that the performance stability of the AOS is much better than that of the FOS. We believe that the proposed design approach is relatively useful and effective as well as can meet the desired goal well. It may provide a new and flexible technique for designing the AOS to match various practical applications in nonlinear optics.

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